EXPERIMENT 6

```
clc;
clear;
close all;

figure('Name','Microwave Semiconductor Devices and Mixer Analysis', ...
    'NumberTitle','off','Color','w', ...
    'Position',[50 50 1350 700]);

sgtitle('Microwave Semiconductor Devices and Mixer Analysis', ...
     'FontSize',15,'FontWeight','bold');

%% 1. Gunn Diode I-V Characteristics
subplot(3,3,1);

Vg = [0 1 2 3 4 5 6 7 8 9 10];
Ig = [0 5 10 15 13 11 9 10.5 12 13.5 15];

plot(Vg,Ig,'b-o','LineWidth',1.8,'MarkerSize',3);
grid on;
xlabel('Voltage (V)');
ylabel('Current (mA)');
title('Gunn Diode I-V Characteristics');
xlim([0 10]);
ylim([0 17]);

%% 2. Schottky Diode I-V Characteristics
subplot(3,3,2);

Vs = linspace(-0.3,0.6,500);
Is = zeros(size(Vs));
```

```
for k = 1:length(Vs)
    if Vs(k) < 0
        Is(k) = 2e-10;
    else
        Is(k) = 1e-10*(exp(Vs(k)/0.06)-1);
    end
end

Is = max(abs(Is),1e-12);

semilogy(Vs,Is,'r','LineWidth',1.8);
grid on;
xlabel('Voltage (V)');
ylabel('Current (A)');
title('Schottky Diode I-V Characteristics');
xlim([-0.3 0.6]);

%% 3. PIN Diode Frequency Response
subplot(3,3,3);

f = linspace(0,20,300);
fc = 5;
response = -10*log10(1+(f/fc).^2);

plot(f,response,'m','LineWidth',1.8);
hold on;
xline(fc,'k--','Cutoff Frequency');
grid on;
xlabel('Frequency (GHz)');
ylabel('Response (dB)');
title('PIN Diode Frequency Response');
```

```
xlim([0 20]);
ylim([-15 0]);

%% 4. Mixer IF Output Spectrum
subplot(3,3,4);

fIF = 0.5;
PIF = -10;

stem(fIF,PIF,'b','LineWidth',1.8, ...
    'Marker','o','MarkerFaceColor','b');
hold on;
plot(fIF,-10,'ro','MarkerFaceColor','r');
grid on;
xlabel('Frequency (GHz)');
ylabel('Power (dBm)');
title('Mixer IF Output Spectrum');
text(fIF+0.03,-25,'Theoretical IF');
xlim([0 1.5]);
ylim([-160 0]);

%% 5. Conversion Loss
subplot(3,3,5);

conversionLoss = 8;
bar(1,conversionLoss,0.55,'FaceColor',[0 0.45 0.74]);
hold on;
yline(6,'g--','6 dB');
yline(10,'r--','10 dB');
grid on;
xlabel('Simulated');
ylabel('Conversion Loss (dB)');
```

```
title('Conversion Loss = 8.00 dB');
xlim([0.4 1.6]);
ylim([0 10]);

%% 6. Effect of LO Power
subplot(3,3,6);

LO = 0:1:13;
CL = [12 11.5 11 10 9.5 9 8.5 8 7.8 7.6 7.5 7.5 7.5 7.5];

plot(LO,CL,'m-o','LineWidth',1.5, ...
    'MarkerSize',4,'MarkerFaceColor','m');
hold on;
plot(7,8,'ys','MarkerSize',7,'MarkerFaceColor','y');
grid on;
xlabel('LO Power (dBm)');
ylabel('Conversion Loss (dB)');
title('Effect of LO Power');
legend('Illustrative model','Operating point', ...
      'Location','northeast');
xlim([0 14]);
ylim([7 13]);

%% 7. Effect of RF Input Power
subplot(3,3,7);

RF = -20:2:0;
IFout = RF - 8;

plot(RF,IFout,'b-o','LineWidth',1.5, ...
    'MarkerSize',4);
hold on;
```

```
plot(-10,-18,'rs','MarkerSize',7, ...
    'MarkerFaceColor','r');
grid on;
xlabel('RF Input Power (dBm)');
ylabel('IF Output Power (dBm)');
title('Effect of RF Input Power');
legend('Illustrative model','Operating point', ...
      'Location','southeast');
xlim([-20 0]);
ylim([-30 -5]);

%% 8. Mixer Frequency Components
subplot(3,3,8);

freq = [0.2 0.5 0.7];
power = [-17 -5 -25];

stem(freq,power,'b','LineWidth',1.5, ...
    'Marker','o','MarkerFaceColor',[0 0.45 0.74]);
grid on;
set(gca,'XTick',freq);
set(gca,'XTickLabel',{'IF','LO','Sum'});
xlabel('Frequency Components');
ylabel('Power (dBm)');
title('Mixer Frequency Components');
xlim([0 0.9]);
ylim([-30 10]);

%% 9. RF-to-IF Power Relationship
subplot(3,3,9);

RF2 = -20:1:0;
```

```
IF2 = RF2 - 8;

plot(RF2,IF2,'b','LineWidth',1.8);
hold on;
yline(-18,'r--','Nominal IF output');
plot(-10,-18,'ko','MarkerFaceColor','y', ...
    'MarkerSize',6);
grid on;
xlabel('RF Input Power (dBm)');
ylabel('IF Output Power (dBm)');
title('RF-to-IF Power Relationship');
xlim([-20 0]);
ylim([-30 -5]);

%% Overall Figure Formatting
set(findall(gcf,'Type','axes'), ...
   'FontSize',8,'LineWidth',0.8);

sgtitle('Microwave Semiconductor Devices and Mixer Analysis', ...
       'FontSize',14,'FontWeight','bold');

% Save the figure
saveas(gcf,'Microwave_Semiconductor_Mixer_Analysis.png');
```

Microwave Semiconductor Devices and Mixer Analysis





